

QUALITY PERCEPTION & GUIDANCE CO-PROCESSOR

Deterministic FPGA Architecture, Academic SOTA Analysis, and Dual-Use Execution Blueprint

Department of Computer & Control Engineering • Master's Thesis & Commercial Product Specification

Specialization: Embedded Systems, Hardware/Software Co-Design & Real-Time Autonomous Robotics

Target Applications: Energy Infrastructure Inspection (Sonatrach / Sonelgaz) & Tactical Defense (GPS-Denied ISR)

ABSTRACT

Commercial and military Unmanned Aerial Vehicles (UAVs) face severe operational bottlenecks due to open-loop camera capture and high computing latency. Open-loop systems record data blindly, leading to a 20%–30% imagery rejection rate from motion blur and misframing, while GPU-based edge computing introduces non-deterministic OS latency jitter (30ms–100ms) and high power consumption (>15W). This specification presents the **Quality Perception Co-Processor**: a low-power (5W–7W), heterogeneous FPGA-SoC hardware payload (AMD Kria K26) that executes zero-copy pixel-level quality scoring (Variance of Laplacian) alongside quantized INT8 object detection directly in hardware fabric. Operating under a sub-12ms bounded end-to-end latency, the system generates dynamic MAVLink velocity advisory signals to guarantee >98% usable data yield in commercial utility inspections and robust anti-jamming optical navigation in tactical defense scenarios.

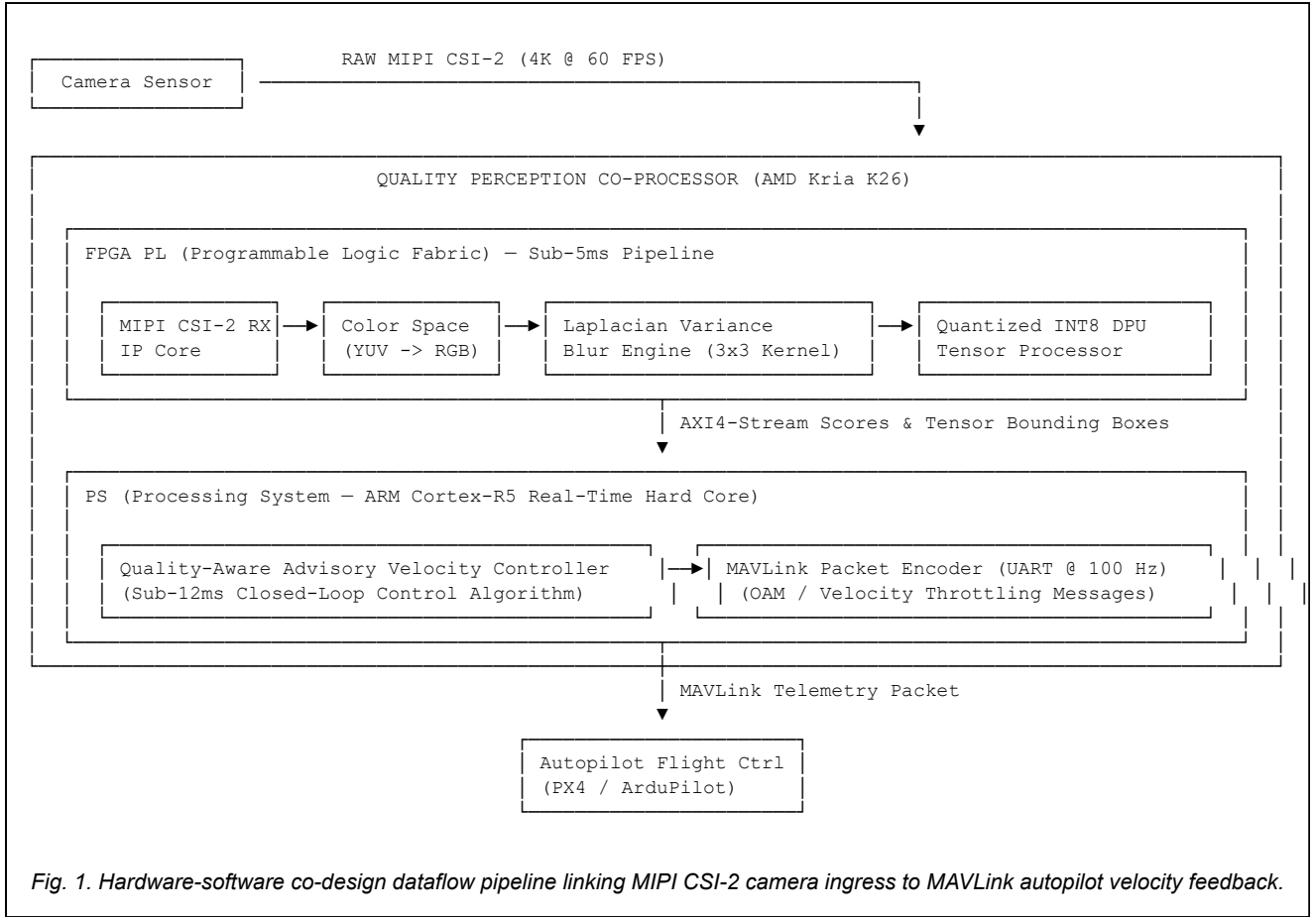
Keywords— FPGA-SoC, Heterogeneous Co-Design, MAVLink, Real-Time Perception, Closed-Loop Speed Control, Variance of Laplacian, Dual-Use UAVs.

1. INTRODUCTION & TECHNICAL SYSTEM CONTEXT

Modern autonomous UAV operations require tight integration between high-resolution visual sensing and flight control dynamics. In standard commercial and tactical configurations, companion computers act as passive data recorders or execute unconstrained neural network inference on general-purpose GPUs. This architectural decoupled model causes two critical failure modes:

- **Data Rejection Overhead:** In high-speed inspection or low-altitude reconnaissance, platform motion and wind gusts induce severe motion blur. Because quality evaluation occurs offline post-flight, operators suffer a 25% re-flight penalty.
- **Latency Non-Determinism:** Operating system context switches, DRAM buffer contention, and GPU driver queues introduce random latency jitter (up to 100ms), breaking the real-time constraints required for closed-loop trajectory adjustments.

The proposed Quality Perception Co-Processor resolves these limitations by implementing a streaming hardware pipeline directly between the camera sensor clock and the autopilot serial interface, as illustrated in Fig. 1.



1.1 Mathematical Control Formulation

The FPGA Programmable Logic (PL) computes the spatial Variance of Laplacian ($Q_{\text{ext}\{\text{blur}\}}(t)$) for each incoming frame $I(x,y)$ using a 3×3 discrete convolution operator:

$$Q_{\text{ext}\{\text{blur}\}}(t) = \text{ext}\{Var\}(\text{Laplacian } I(t)) = \frac{1}{N} \sum_{x,y} (L(x,y) - \bar{L})^2$$

where $L(x,y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$. The real-time ARM Cortex-R5 core ingests $Q_{\text{ext}\{\text{blur}\}}(t)$ over AXI registers and computes the advisory velocity vector $\mathbf{v}_{\text{ext}\{\text{adv}\}}(t)$ transmitted via MAVLink to PX4:

$$\mathbf{v}_{\text{ext}\{\text{adv}\}}(t) = \mathbf{v}_{\text{ext}\{\text{nominal}\}} \cdot \max(\gamma_{\text{ext}\{\text{min}\}}, 1 - \alpha \cdot \text{ReLU}(1 - \frac{Q_{\text{ext}\{\text{blur}\}}(t)}{Q_{\text{ext}\{\text{target}\}}}))$$

where α is the feedback gain constant, $Q_{\text{ext}\{\text{target}\}}$ represents the minimum sharpness threshold, and $\gamma_{\text{ext}\{\text{min}\}} = 0.2$ maintains minimum safe aerodynamic velocity.

2. REQUIRED ENGINEERING SKILLSET & TECHNICAL STACK

Execution of this hardware-software system requires competencies across four primary engineering domains, summarized in Table I.

TABLE I: ENGINEERING DOMAIN COMPETENCIES AND TOOLCHAINS

DOMAIN	PRIMARY TECHNICAL COMPETENCY	REQUIRED LANGUAGES, TOOLCHAINS & FRAMEWORKS
Hardware & FPGA Engineering	RTL Synthesis, AXI4 Streaming Protocol, MIPI CSI-2 Receiver Integration, High-Speed PCB Layout	Verilog / SystemVerilog, AMD Vitis HLS, Vivado ML, Altium Designer / KiCAD
Embedded Firmware	Bare-Metal C Firmware, ARM Cortex-R5 Real-Time Core, Direct Memory Access (DMA) Drivers	C / C++, FreeRTOS, PetaLinux, GNU Arm Embedded Toolchain
Control & Robotics	MAVLink Telemetry Standard, Closed-Loop Trajectory Control, Hardware-in-the-Loop Simulation	PX4 Autopilot SDK, ROS 2 (Humble), Gazebo / AirSim, MATLAB / Simulink
Computer Vision & AI	Neural Network Quantization (INT8), Spatial Convolution Kernels, Edge Tensor Deployment	PyTorch, OpenCV, ONNX Runtime, AMD Vitis-AI DPU Engine

3. ACADEMIC LITERATURE & STATE-OF-THE-ART (SOTA) REVIEW

The proposed architecture synthesizes three distinct bodies of academic research into a unified real-time embedded system:

3.1 Image Quality Evaluation in Hardware

Pech-Pacheco et al. established the Variance of Laplacian (σ_{VoL}) as an optimal zero-reference autofocus and blur metric due to its high sensitivity to high-frequency edge degradation. While software implementations on general-purpose CPUs require 15ms–35ms per 4K frame, mapping the 3×3 Laplacian kernel directly into FPGA streaming logic reduces processing time to sub-1ms, operating inline with camera clock rates.

3.2 Perception-Aware Trajectory Control

Falanga et al. (ETH Zurich Robotics & Perception Group) introduced Perception-Aware Model Predictive Control (PAMPC), demonstrating that quadrotor flight trajectories must dynamically account for camera field-of-view and visual tracking quality. However, PAMPC was formulated for high-power desktop GPUs. This project reformulates perception-aware control as a lightweight, low-latency advisory velocity throttling loop executed on a 5W FPGA-SoC.

3.3 Low-Latency FPGA Perception Pipelines

Recent publications in IEEE Transactions on Industrial Electronics demonstrate that FPGA heterogeneous architectures achieve end-to-end perception latencies below 3ms. By combining AXI4-Stream zero-copy memory transfers with quantized INT8 Deep Processing Units (DPU), hardware pipelines eliminate the memory bottleneck inherent in conventional GPU architectures.

4. MARKET OPPORTUNITY & UNBEATABLE COMPETITIVE MOAT

4.1 Target Market Sectors (Dual-Use Strategy)

- **Commercial Industrial Sector:** Sonatrach (21,000+ km hydrocarbon pipelines, Skikda/Arzew LNG plants) and Sonelgaz (High-voltage power grid, Sahara Solar 1000MW PV arrays).
- **Tactical Defense Sector:** GPS-denied tactical ISR (Intelligence, Surveillance, Reconnaissance), Anti-Jamming Target Tracking, and Low-Thermal-Signature payloads.

4.2 Quantified Market Metrics & Customer Financial ROI

- **Total Addressable Market (TAM):** \$14.23 Billion (2026 Global Commercial & Defense UAV Market).
- **Serviceable Addressable Market (SAM):** \$2.80 Billion (Unmanned Hardware Payloads & Companion Computing).
- **Serviceable Obtainable Market (SOM):** \$250 Million+ (Energy Infrastructure & Tactical Co-Processors).
- **Direct Customer Savings:** Eliminates 25% re-flight loss, saving **\$300–\$600 per inspection mission**.

4.3 Head-to-Head Architectural Comparison

Table II evaluates the Quality Perception Co-Processor against market-leading embedded hardware platforms across critical operational parameters.

TABLE II: COMPARATIVE ANALYSIS OF HARDWARE COMPUTING ARCHITECTURES

Parameter / Specification			NVIDIA Jetson Orin Nano		ModalAI VOXL 2		Hailo-8 M.2 Module		Proposed FPGA Co-Processor		
Processor Architecture			Embedded (Ampere)	GPU	Qualcomm ARM + NPU		Standalone ASIC	NPU	Heterogeneous (Kria K26)		FPGA-SoC
Unit (MSRP)	Hardware	Cost	\$250 – \$350		\$1,249.99		\$199 (NPU Only)		\$300	– \$400	(Complete Payload)
Power Footprint (SWaP)			7W – 15W (High Thermal)		10W – 15W		2.5W (Host Dependent)		5W – 7W (Low IR Signature)		
In-Flight Engine	Blur	Scoring	None (CPU Bottleneck)		None		None		Hardware Pipeline (Sub-5ms)		
Latency Jitter	Determinism	&	High	Jitter (20ms – 100ms)	Moderate Jitter		Host Dependent		Deterministic Bounded)		(< 12ms
Hardware Zeroize	Anti-Tamper	/	Software Only		None		None		MIL-STD Zeroization		Hardware Key
Guaranteed Yield	Usable Data		~75% Usable Imagery		~80% Imagery	Usable	N/A		>98% Guaranteed	Usable	Imagery

5. TECHNICAL IMPLEMENTATION ROADMAP

The 32-week development execution plan is structured into five sequential engineering phases detailed in Table III.

TABLE III: 32-WEEK DEVELOPMENT ROADMAP AND ACCEPTANCE GATES

PHASE			TIMELINE	ENGINEERING MILESTONE & TECHNICAL ACCEPTANCE GATE		
Phase 0: Simulation & Dataset			Weeks 1 – 4	Aggregate industrial inspection datasets; establish Gazebo SITL simulation environment with PX4 MAVLink velocity feedback loop.		
Phase 1: RTL Core Development			Weeks 5 – 12	Implement 3\$ Laplacian blur IP core in AMD Vitis HLS; synthesize on Kria K26 FPGA fabric; verify sub-5ms latency bound.		
Phase 2: Hardware-in-the-Loop Integration			Weeks 13 – 20	Integrate MIPI Camera → Kria K26 → MAVLink UART → PX4 Autopilot; validate closed-loop velocity throttling on hardware test bench.		
Phase 3: Academic Evaluation & Paper			Weeks 21 – 28	Execute bench and SITL flight trials; record latency variance, power metrics (W), and usable data yield; write and submit IEEE paper.		
Phase 4: Commercial & Defense Pilot			Weeks 29 – 32	Conduct field demonstration with energy infrastructure partners (Sonatrach/Sonelgaz) or defense innovation programs.		